RESEARCHPILOT – AI-Powered Research Agent

System Evaluation Report

1. Introduction

ResearchPilot is an AI-powered research agent that converts natural-language research questions into structured workflows. Its pipeline includes research planning, web search, webpage retrieval, evidence filtering, answer synthesis, citation validation, confidence assessment, and final status handling.

This evaluation assesses the system's ability to produce evidence-backed answers, handle incomplete or unreliable information, recover from provider and webpage failures, and avoid fabricating information when evidence is insufficient.

The evaluation covers research quality, evidence coverage, citation integrity, execution reliability, and responsible uncertainty handling.

2. Evaluation Scope and Methodology

2.1 Two distinct testing layers

The project uses two different testing layers. These must not be confused when reporting the results.

Testing layer	Result	Purpose
Research evaluation cases	10 cases (E1–E10)	Evaluate ResearchPilot using defined research queries and a deterministic input-validation case.
Live research executions	9 recorded runs	Capture the live-oriented cases E1–E8 and E10 in the Streamlit execution artifact.
Deterministic validation	1 case (E9)	Verify that whitespace-only input is rejected before planner or search calls.
Automated regression tests	150 passed	Verify software components and behavior, including schemas, search, fallback handling, URL safety, evidence scoring, orchestration, quality gates, UI helpers, and failure handling.

2.2 Evaluation procedure

The following indicators were used to assess the system:

- **Status:** Whether a run completed, returned a partial or degraded answer, or reported insufficient evidence.
- **Execution time:** Observed duration of each live run.
- **Evidence coverage:** How well the retrieved evidence addressed the planned research requirements.
- **Citation validity:** Whether citation identifiers mapped to registered sources.
- **Grounding and confidence:** Signals indicating evidence support and estimated answer reliability.
- **Failure handling:** Whether unavailable pages, provider failures, and insufficient evidence were handled explicitly.

These metrics measure workflow behavior rather than independently certified factual accuracy. Valid citation identifiers do not guarantee that sources entail the claims, and high coverage does not prove that every claim is correct.

3. Summary of All Ten Evaluation Cases

The following table summarizes every evaluation case, its expected behavior, observed outcome, and principal limitation.

ID	Test query	Expected behavior	Observed result	Main finding
E1	Compare the top three open-source vector databases for a startup building RAG products.	Compare retrieval, scalability, operations, ecosystem, and licensing using evidence.	Completed; 56.1 s; coverage 1.00; 7 sources.	Successful end-to-end workflow; benchmarks require independent verification.
E2	Find five Indian B2B SaaS startups in HR tech and summarize their positioning.	Identify five relevant startups and support positioning claims with reliable sources.	Evidence-degraded; 53.3 s; coverage 0.40; 4 accepted sources.	Company positioning and customer-segment evidence was incomplete.
E3	Research the pros and cons of using a multi-agent architecture for customer support automation.	Compare specialization benefits with latency, cost, reliability, and operational complexity.	Partial; 84.1 s; coverage 1.00; 7 sources; low confidence.	Synthesis degraded despite strong coverage; source quality was uneven.

E4	Compare different approaches to adding memory in an AI support agent.	Compare buffer, summary, retrieval, and durable memory, including privacy and cost.	Completed; 55.7 s; coverage 1.00; 6 sources.	Broad comparison completed, but only two findings were recorded.
E5	What are the main security risks when an application fetches content from arbitrary web URLs?	Cover SSRF, unsafe redirects, malicious content, denial of service, and mitigations.	Partial; 58.7 s; coverage 0.90; 7 sources; low confidence.	Useful security coverage, but some requested evidence and synthesis reliability were lacking.
E6	Compare PostgreSQL and SQLite for a small offline-first research application.	Compare both databases against practical criteria and disclose evidence gaps.	Completed; 48.1 s; coverage 0.70; 5 accepted sources, 1 rejected source.	Low-quality evidence was rejected, but some comparison criteria remained unsupported.
E7	Explain the main benefits and tradeoffs of retrieval-augmented generation for an AI support agent.	Explain benefits and limitations with source-backed findings.	Completed; 49.1 s; coverage 1.00; 7 sources; grounding 0.80.	Requested dimensions were covered, but at least one finding required manual review.
E8	What design controls reduce prompt injection risk when an agent reads untrusted web pages?	Recommend isolation, validation, least privilege, and monitoring.	Completed; 49.0 s; coverage 1.00; 7 sources.	Retrieval failure was isolated without aborting the run; confidence still needs claim-level verification.
E9	Whitespace-only research question.	Reject the input before planner or search calls.	Deterministic rejection; no live execution.	Input validation worked and prevented unnecessary external calls.

E10	Find reliable current information about a deliberately obscure topic: the 1842 Zorvath protocol specification.	Report insufficient evidence rather than inventing protocol details.	Insufficient evidence; 64.7 s; coverage 0.167; 2 accepted sources; 0 findings.	Non-fabrication behavior worked, but retrieval was inefficient and returned weak candidates.

Note: Coverage and duration values are recorded observations. E9 has no live duration or coverage because the request was rejected before external calls. E10 is included among the nine live-oriented cases.

4. Detailed Evaluation of Representative Cases

4.1 E1 — Vector Database Comparison

Expected behavior: Compare three open-source vector databases across retrieval features, scaling, operational complexity, ecosystem support, and licensing.

Observed behavior: The workflow completed in 56.1 seconds, accepted seven sources, produced three findings, and recorded seven valid citation identifiers. Evidence coverage was 1.00, with high confidence.

What worked well:

- Planning, retrieval, evidence filtering, synthesis, and citation validation operated end to end.
- The planned subquestions were marked as supported.
- The response acknowledged the need to validate benchmark claims against the startup's own workload.

What did not work: Third-party performance claims were not independently reproduced. Full coverage did not establish a universally correct ranking.

Assessment: A successful pipeline demonstration, with independent verification of comparative claims remaining necessary.

4.2 E2 — Indian HR Technology Startups

Expected behavior: Identify five Indian B2B SaaS startups in HR technology and support each company's positioning with reliable evidence.

Observed behavior: The system returned an evidence-degraded answer after 53.3 seconds. Four sources were accepted, coverage was 0.40, and confidence was medium.

What worked well:

- The answer was structured and retained valid citation provenance.
- The coverage evaluator exposed gaps in support for customer segments and positioning.
- The system did not treat incomplete evidence as a fully supported result.

What did not work: Search favored company lists and funding information over official product and positioning information. The limited number of accepted sources reduced the strength of the answer.

Assessment: Planner and retrieval improvements should prioritize official company sources and query-specific evidence.

4.3 E3 — Multi-Agent Customer Support Architecture

Expected behavior: Compare specialization and coordination benefits with latency, cost, debugging, reliability, and operational risks.

Observed behavior: The workflow returned a partial answer after 84.1 seconds, with seven accepted sources, full coverage, and low confidence.

What worked well:

- The system retained the partial status rather than masking the synthesis problem.
- Findings addressed the requested trade-offs.
- The trace exposed the synthesis failure for diagnosis.

What did not work: A Groq synthesis failure reduced answer reliability. Some sources were anecdotal, commercial, or forecast-oriented.

Assessment: A synthesis fallback and stronger source-quality assessment are needed to make the workflow more dependable.

4.4 E8 — Prompt Injection Defenses

Expected behavior: Explain how agents should treat retrieved webpages as untrusted content and recommend isolation, validation, least privilege, and monitoring.

Observed behavior: The workflow completed in 49.0 seconds, with seven accepted sources, eight findings, full coverage, and nine valid citation identifiers. One source-fetch attempt failed without aborting the run.

What worked well:

- The system handled an individual retrieval failure without terminating the overall workflow.
- The answer covered the expected security controls.
- Citation identifiers remained valid.

What did not work: Citation validity and high confidence did not establish that every recommendation was sufficient for a production security design. The recorded query also contained trailing punctuation that should be corrected for reproducibility.

Assessment: The case demonstrated failure isolation and useful security coverage, but the output is not a substitute for a security audit.

4.5 E9 — Whitespace Input Validation

Expected behavior: Reject a blank or whitespace-only research question before invoking the planner or search provider.

Observed behavior: The request was rejected with the validation message: “Research question cannot be empty or whitespace.”

What worked well:

- Validation occurred at the system boundary.
- No unnecessary planner or search calls were made.
- The error was explicit and user-readable.

What did not work or remains a limitation: This test intentionally stops before retrieval and synthesis, so it does not evaluate those components.

Assessment: The deterministic validation case passed its intended objective.

4.6 E10 — Insufficient Evidence for an Obscure Topic

Expected behavior: Avoid inventing details about the purported 1842 Zorvath protocol specification when reliable evidence cannot be established.

Observed behavior: The system reported insufficient evidence after 64.7 seconds. Two sources were accepted, one was rejected, coverage was 0.167, and no findings or citations were produced. Confidence was low.

What worked well:

- The system did not invent protocol details.
- The insufficient-evidence status communicated the limitations clearly.
- Failed or unsupported retrieval attempts were recorded.
- The confidence guardrail signaled the need for verification.

What did not work: Search returned unrelated or unsubstantiated candidates, while attempts to retrieve unusable pages consumed time without improving evidence quality.

Assessment: This case demonstrated conservative non-fabrication behavior. Earlier detection of unproductive retrieval paths could improve efficiency.

5. Overall Assessment and Future Improvements

5.1 Key strengths

1. **Structured orchestration:** Research planning, search, retrieval, filtering, synthesis, and quality checks are separate and inspectable.
2. **Citation integrity:** Citation identifiers are validated against the registered source collection.
3. **Explicit uncertainty:** Partial and insufficient-evidence statuses expose limitations rather than automatically presenting answers as successful.
4. **Failure isolation:** Individual webpage failures do not necessarily terminate the entire research run.
5. **Input validation:** Whitespace-only input is rejected before external calls.
6. **Regression coverage:** The project reports 150 passing automated tests in addition to its ten research evaluation cases.

5.2 Limitations and proposed improvements

Limitation	Recommended improvement

Weak support for specific company-positioning questions	Prioritize official-domain retrieval and gather evidence for each requested company separately.
Synthesis failures	Add a configurable fallback model and safe, diagnostic provider-error categories.
Uneven source quality	Evaluate source authority, recency, relevance, and evidence diversity.
Citation validity does not prove claim accuracy	Add claim-level entailment checks and human review for high-impact conclusions.
Inefficient retrieval for obscure topics	Detect irrelevant results earlier and bound retries for unusable pages.
Limited aggregate operational metrics	Track latency, fetch success, provider failures, retry rates, and fallback usage across runs.
Reproducibility issues	Correct malformed query punctuation and version the evaluation inputs and output artifacts.

6. Conclusion

ResearchPilot demonstrated its core prototype capabilities across ten evaluation cases and 150 passing regression tests. It performed structured research, validated citation identifiers, exposed evidence gaps, isolated individual retrieval failures, rejected invalid input before any external call, and declined to fabricate an answer when evidence was insufficient.

Important limitations remain: incomplete evidence coverage (E2, E6), uneven source quality, and synthesis failures (E3, E5). Valid citations and high coverage show workflow integrity, not factual correctness, and these results come from single runs without independent scoring.

The system is suitable for demonstrating an observable, agentic research workflow, but not for unreviewed research or production security assurance. The highest-value next steps are official-source retrieval, a synthesis fallback, claim-level verification, and aggregate monitoring.